

SANKALP SAMARTH

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Education

Manipal Institute of Technology (1st Year Undergraduate Student)

Jul 2025 – Jul 2029

Bachelor of Technology - Computer Science and Engineering

Bengaluru, Karnataka

Relevant Coursework: Object Oriented Programming (Java), Data Visualizations (Pandas, Numpy, Matplotlib), Artificial Intelligence, Software Engineering

Kerala Samajam Model School, Jamshedpur: Class 12th: ISC (2025) - 94.25%, Class 10th: ICSE (2023) - 95.2%

Technical Skills

- **Languages:** Java, Python, Typescript, C, HTML/CSS
- **Databases:** MongoDB, Supabase
- **Frameworks and Libraries:** Pandas, Numpy, Matplotlib, TensorFlow, PyTorch, scikit-learn, Selenium
- **Tools:** Jupyter Notebook, Google Colab, VS Code, Git, GitHub, Figma, Notion
- **Soft Skills:** Research, Analysis, Innovation, Critical Thinking, Leadership, Management, Effective Communication

Experience

NASA Human Exploration Rover Challenge (International)

Aug 2025 – Sep 2025

Research Intern

NASA's MSFC, Huntsville, USA

- During this period, the team was instructed to implement innovative engineering solutions: switchable two-wheel/four-wheel drive, folding mechanism to meet volume constraints, enhancing rover performance and completing the obstacle course while increasing team efficiency by 15%.

Projects

Spanda Life: Full-Stack Wellness Retreat Booking & E-Commerce Platform ([Repo Link](#))

Mar 2026

- Developed a production-ready wellness platform with headless Shopify integration, structured booking workflows, real-time cart synchronization, secure backend handling, and CMS-enabled content management for a non-technical client.
- **Tech Stack:** React, TypeScript, Tailwind CSS, Shopify Storefront API, Supabase, PostgreSQL, Deno Edge Functions, Zustand

Parkingo: IOT-based Real Time Parking Management System

Sep 2024

- Built an IoT-enabled smart parking system to automate vehicle entry/exit, real-time slot monitoring, and intelligent parking workflows using RFID, IR sensors, ESP32-CAM, and multi-microcontroller communication for vehicle identification, barrier control, and QR-based parking management.
- Developed a GSM-enabled digital parking ecosystem featuring automated parking receipts, slot details, timestamps, billing notifications, and SMS-based user alerts, while conducting a field study at a 320-slot mall parking facility in Jamshedpur to design a faster, paperless, and eco-friendly parking solution.
- **Tech Stack:**
 - Software/Technologies:* Embedded C/C++ (Arduino IDE), UART Serial Communication, IoT Automation, QR Code Integration, Sensor Interfacing, GSM Communication
 - Hardware:* Arduino Mega, ESP32-CAM, ESP8266/NodeMCU, RFID Module, GSM Module, IR Sensors, Servo Motor, TFT/OLED Displays

Awards, Certifications & Publications

Secured **1st Position at NIT Jamshedpur** and **2nd Position at IIT Dhanbad** for developing “**Parkingo**”, an innovative smart parking automation system.